Hidden Debt Revelation and Optimal Default

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Abstract

Hidden public liabilities are a pervasive but poorly measured feature of sovereign balance sheets. When they are eventually revealed, they can reprice sovereign risk overnight and compress the policy space between painful fiscal adjustment and default. This paper studies optimal sovereign default following the revelation of hidden debt. We develop a quantitative model of optimal sovereign default following a hidden-debt revelation and apply it to Senegal's 2024 shock, in which an incoming administration uncovered more than ten billion dollars in off-balance-sheet obligations, revising central-government debt upward by 50 percentage points of GDP. We embed the revelation as a one-time upward shift in the initial debt stock within a small open economy model with long-term bonds, downward nominal wage rigidity, and output costs calibrated to Senegal's monetary union environment. The baseline model predicts that default would have been optimal from 2023 onward, once the retroactively corrected debt-to-GDP ratio crossed the default threshold of 107 percent. Since Senegal has not defaulted as of early 2026, we provide conditions that rationalize continued repayment with income dynamics that are consistent with either optimism about mean reversion or gambling for redemption. Under this no-default calibration, the country is solvent but fragile: an adverse income shock of 3 percent would lead to an expected time to default of less than one year. Both calibrations point to the same conclusion: Senegal moved from a position of moderate fiscal risk into a regime of acute vulnerability after the revelation of its hidden debt.

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1 Introduction

In March 2024, a newly elected Senegalese government commissioned an audit of its fiscal accounts and discovered that its predecessor had accumulated more than ten billion dollars in liabilities outside official statistics. Central-government debt, reported at roughly 70 percent of GDP, was revised upward by 50 percentage points overnight. The IMF suspended its program. Rating agencies downgraded Senegal's sovereign credit. Eurobond spreads surged into distressed territory. Yet the authorities declined to restructure, invoking the sovereign's "right" to decide whether and how to address the debt problem. This episode is an extreme realization of a broader phenomenon: hidden public liabilities that compress the policy space between fiscal adjustment and default the moment they are revealed. *When is it optimal for a government facing a large hidden-debt revelation and very high spreads to default rather than repay?*

Governments routinely promise more than they record on their official balance sheets. Off-budget borrowing, guarantees, and arrears allow them to shift expenditures across time and to soften political constraints, but they also create hidden liabilities that are only imperfectly understood by markets and international institutions. Standard models of sovereign default (Eaton and Gersovitz, 1981; Arellano, 2008) assume that public debt is fully observable and that spreads reflect fundamentals in real time. In practice, much of the variation in spreads around crises is driven by news about the stock of obligations that were previously unmeasured or misreported (Horn et al., 2021). Hidden debt is therefore not just a measurement error; it is a state variable that shapes incentives to default.

This paper develops a quantitative model of optimal sovereign default following a hidden-debt revelation and applies it to Senegal's 2024 shock. We model the revelation as an initial condition: an audit discloses obligations excluded from official statistics, and the government inherits the enlarged debt stock. Senegal's membership in the West African Economic and Monetary Union (WAEMU) sharpens the tradeoff considerably — the fixed CFA franc peg rules out devaluation and monetary financing, so the entire burden of adjustment falls on fiscal policy or restructuring.

Our main finding is that, under the income process estimated from Senegal's tradable sector, default would have been the optimal policy from 2023 onward. The hidden-debt revelation alone was sufficient to push the corrected debt-to-GDP ratio past the model's default threshold of 107 percent — a crossing that occurred between 2022 and 2023, before the audit was even conducted. Since Senegal has not defaulted as of early 2026, we also characterize the conditions under which continued repayment is rational. A no-default calibration that matches post-revelation spread dynamics requires market-implied income shocks that are more volatile and less persistent than historical data suggests, reflecting either optimism about mean reversion, perhaps driven by anticipated oil revenues and governance reforms, or a form of gambling for redemption in which the government delays restructuring in the hope that favorable shocks restore sustainability. Under this calibration, Senegal is solvent but fragile: an adverse income shock of 3 percent would lead to an expected time to default of less than one year.

We obtain these findings in a small open economy model with long-term bonds, CES preferences over tradable and non-tradable goods, downward nominal wage rigidity, and output costs disciplined by Senegal's monetary union environment, and building on Arellano (2008), Chatterjee and Eyigungor (2012), and Bianchi and Sosa-Padilla (2024). Hidden debt enters as a one-time upward shift in the initial debt stock. The tradable income process is estimated from World Development Indicators data on Senegal's agriculture and industry value added over 1980–2024, and the model is calibrated to match the pre-revelation Senegalese government bond risk premium observations, proxied by the J.P. Morgan Emerging Markets Bond Index Global (EMBIG) for Senegal, at officially reported debt. However, because the hidden liabilities existed throughout the pre-revelation period, those spreads may already embed non-public information about the true debt stock — in which case the reported debt level is not the state variable markets were actually pricing. To address this, our no-default calibration instead holds debt fixed at the corrected post-revelation level and exploits variation in real income and EMBIG spreads to jointly identify the output cost and the market-implied income process. Both calibrations point to the same conclusion: Senegal moved from a position of moderate fiscal risk into a regime of acute vulnerability after the revelation of its hidden debt.

The rest of the paper is organized as follows. Section 2 documents the Senegalese debt shock and the evolution of sovereign spreads before and after the revelation. Section 3 develops the model and defines the recursive equilibrium. Section 4 presents the calibration and quantitative results. Section 5 concludes.

2 Motivation and Background

2.1 Hidden Sovereign Debt Revelations

Hidden public debt arises when governments take on obligations that are not transparently recorded in headline debt statistics. Common channels include borrowing by public enterprises with implicit guarantees, off–budget funds, rolling bank loans, and the systematic understatement of fiscal deficits (Horn et al., 2021). These practices can be politically attractive in the short run: they allow governments to finance spending surges or election–year transfers while appearing to respect formal fiscal rules.

Economically, however, hidden debt has two key consequences. First, it breaks the link between observed debt ratios and the true intertemporal budget constraint. Investors and international institutions may underestimate a country's vulnerability, leading to excessively low spreads in tranquil times. Second, the eventual revelation of hidden liabilities can be a large negative shock to the perceived net worth of the sovereign. When revelation occurs, spreads jump, and the government faces a compressed menu of painful choices: tighten policy aggressively, attempt to grow out of the imbalance, or restructure and default.

Most sovereign-debt models abstract from this distinction between reported and true debt. In the data, though, episodes in which new administrations or audit institutions uncover large hidden liabilities are common (Horn et al., 2021; Reinhart and Rogoff, 2009), and they are often followed by sharp revisions in debt ratios, rating downgrades, and spread spikes. The Senegalese episode is an extreme realization of this mechanism.

2.2 Senegal's Hidden-Debt Shock

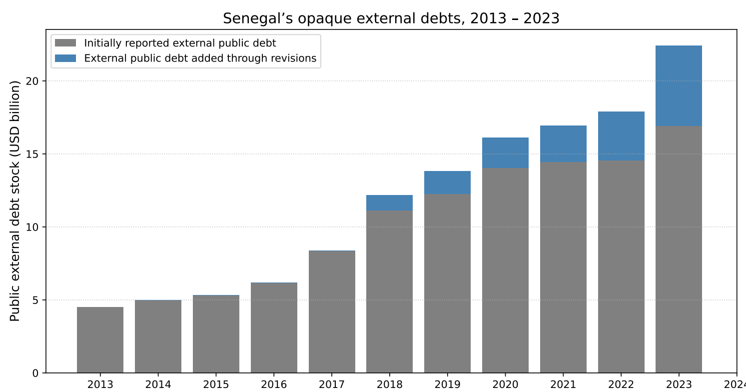
In Senegal, a change of administration in 2024 led to a comprehensive audit of the fiscal accounts and public borrowing. The audit revealed sizable off-budget borrowing through commercial banks, special deposit accounts, and quasi-fiscal operations, and it revised the central government debt-to-GDP ratio from the mid-70s to well above 100 percent. The Court of Accounts and internal reports documented over-financing of the budget, hidden bank loans outside normal procedures, and incomplete recording of a large sukuk issuance. Taken together, these elements imply more than ten billion dollars of hidden liabilities and a debt ratio that climbs into the 110- 120 percent range in the years following the pandemic.

The scale of under-reporting can be quantified by comparing Senegal's debt reports across different vintages of the World Bank's International Debt Statistics. Figure 1 illustrates this pattern for the 2013–2023 period. When initially reported in the 2024 vintage, external public and publicly guaranteed debt appeared relatively stable. However, the 2025 vintage—published after the audit—revealed substantial upward revisions throughout the entire period. The blue bars in Figure 1 show additional debt stocks that were added through revisions, with the gap between initially reported and revised figures widening significantly in recent years. By 2023, several billion dollars of external obligations that should have been reported contemporaneously only appeared in the statistics retrospectively, illustrating the systematic nature of debt under-reporting that preceded the full revelation.

The revision had two immediate macroeconomic consequences. First, the existing IMF program, approved in 2023 on the basis of understated data, was suspended once the scale of misreporting became apparent. Second, rating agencies began to downgrade

Senegal's sovereign rating, and bond markets repriced Senegalese risk sharply upward. The sovereign, which had been borrowing at spreads comparable to other frontier issuers in the region, suddenly found itself priced as a distressed credit. Ndiaye and Kessler (2026) provides further background information and a descriptive analysis of Senegal's hidden debt shock.

The episode is particularly striking because it occurred within the institutional constraints of a monetary union. Senegal is a member of the West African Economic and Monetary Union (WAEMU), which uses a common currency pegged to the euro and administered by a regional central bank. This arrangement anchors inflation but rules out unilateral exchange-rate devaluation and monetary financing (Arellano et al., 2021). A country that experiences a large debt shock in this environment cannot inflate its way out of trouble; the burden of adjustment falls on fiscal policy, structural reforms, and, potentially, debt restructuring. At the same time, the regional banking system holds a large fraction of domestic sovereign paper, so any restructuring that starts with domestic claims risks spilling over into bank capital, deposits, and ultimately the currency peg.



Notes: The figure shows external public debt values for Senegal from the IDS 2024 Vintage and the most recently reported public and publicly guaranteed debt stocks between 2013 and 2023 in billion USD. The gray bars show the initially reported debt stocks in 2024. The blue bars show additional debt stocks added through revisions in 2025. Source: World Bank IDS.

Figure 1: Senegal's Hidden Debt, 2013-2023

Against this backdrop, the Senegalese authorities have emphasized the objective of preserving external market access and avoiding an outright default, at least in the short run. Public statements have emphasized the sovereign's right to decide whether and how to tackle the debt problem and have signaled a reluctance to initiate a restructuring, even as spreads moved into double-digit territory. This policy stance raises a natural question: given a large hidden-debt revelation and very high spreads, when is it optimal for a sovereign to continue servicing its debt, and when is it optimal to restructure?

2.3 Consequences of Senegal's Hidden Debt Revelation

Figure 2 plots a time series of Senegal's sovereign spread over U.S. Treasuries, as measured by an index of hard-currency sovereign bonds, from late 2023 to late 2025. The EMBIG series is reconstructed from secondary-market data and is intended as a proxy for Senegalese government bond's risk premium rather than a definitive measure. Two vertical lines mark key dates: the formal publication of the audit that crystallized the hidden-debt shock, and a later public statement by the authorities signaling that Senegal would not restructure its external obligations.

Figure 2 highlights three features that guide the model. First, spreads were relatively moderate before the revelation, despite pre-existing concerns about deficits and public enterprises, suggesting that markets were not fully pricing the hidden liabilities. Second, spreads jumped sharply once the audit crystallized the scale of hidden obligations and the IMF program was suspended. Third, and most importantly for our purposes, spreads remained elevated even after policymakers publicly committed to repayment and ruled out restructuring — a signal that markets continued to assign substantial default probability despite the government's stated intentions. These three patterns are difficult to reconcile with a model in which debt is always fully observed and default is triggered solely by contemporaneous income shocks. They are, however, natural in a framework where the revelation of hidden liabilities constitutes a discrete upward shift in the debt stock that permanently alters the government's default calculus, independent of any income deterioration.

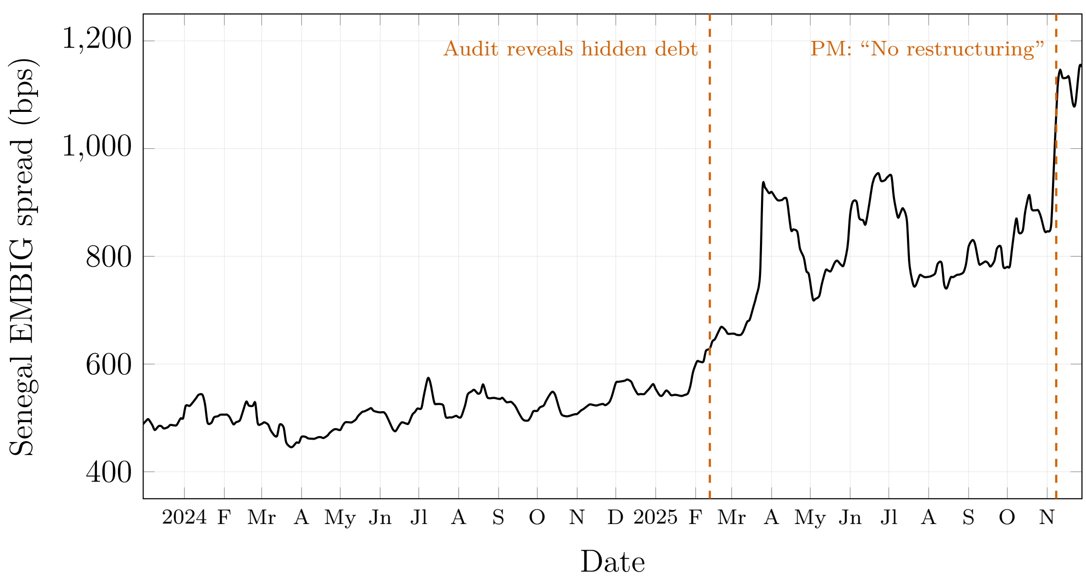


Figure 2: Senegal' Sovereign Spread and Key Events.

3 Model

We consider a small open economy inhabited by a representative household and governed by a benevolent government that lacks commitment over future fiscal policies. The government borrows from a continuum of competitive, risk-neutral international lenders by issuing long-term non-contingent bonds. The model begins immediately after the revelation of hidden liabilities documented in Section 2, so that the hidden debt enters as an initial condition rather than an ongoing state variable. We first describe the economic environment, then formulate the government's problem, the lenders' pricing decisions, and the recursive equilibrium, building on the quantitative sovereign default literature (Arellano, 2008; Chatterjee and Eyigungor, 2012; Bianchi and Sosa-Padilla, 2024).

3.1 Households and Firms

Preferences. The representative agent in the borrowing economy has preferences given by

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t U(c_t), \tag{1}$$

where β denotes the subjective discount factor and c_t represents consumption. The utility function is strictly increasing and concave and takes the standard CRRA form $u(c_t) = (c_t^{1-\sigma} - 1)/(1-\sigma)$ where c_t is a CES composite between tradable and non-tradable consumption goods:

$$c_t = \left[\alpha (c_t^T)^{\frac{\gamma-1}{\gamma}} + (1-\alpha)(c_t^N)^{\frac{\gamma-1}{\gamma}}\right]^{\frac{\gamma}{\gamma-1}} \tag{2}$$

We will use $u(c_t^T, c_t^N)$ to denote the utility as a function of the two consumption goods. Each period, households supply inelastically $\bar{h}$ hours to the labor market. However, due to the presence of downward nominal wage rigidity, households may not be able to sell all of the hours they supply. As a result, total hours worked, $h_t \leq h$, is taken as given by households. Households also receive a stochastic endowment of the tradable good $y^T \in \mathcal{Y}$. The households' budget constraint, expressed in terms of tradables, is thus given by

$$c_t^T + p_t^N c_t^N = y_t^T + \phi_t^N + w_t h_t + T_t, \tag{4}$$

where T_t is the lump-sum transfers from the government p_t^N denotes the price of non-tradables and w_t denotes the wage rate. Assuming that the law of one price holds and that the price of tradable goods in foreign currency is constant and normalized to one, these prices can also be interpreted in terms of foreign currency.

Production. The non-tradable good is produced by a continuum of firms in a perfectly competitive market. Each firm produces a non-tradable good according to a linear production technology given by $y_t^N = h_t$ and obtains profits $\phi_t^N = p_t^N h_t - w_t h_t$. Given the linear production function, we obtain that in equilibrium, $p_t^N = w_t$.

Monetary union and nominal rigidities. Senegal is a member of WAEMU, whose common currency, the CFA franc, is pegged to the euro at a fixed rate.[1] The union's central bank is prohibited from extending credit to member governments beyond narrowly defined statutory advances and cannot devalue to reduce the real value of a member country's external obligations. As a result, we assume that the nominal exchange rate e expressed in units of domestic currency per foreign currency is assumed to be fixed.

Following Schmitt-Grohé and Uribe (2016), we also assume that there exists a minimum wage in nominal terms; i.e., $W_t \geq \overline{W}$. Defining $\overline{W} = \frac{\overline{W}}{e}$, we arrive at $w_t \geq \overline{W}$. Thus, if the market-clearing wage exceeds $\overline{w}$, equilibrium employment equals the aggregate endowment of labor. Otherwise, hours are determined by labor demand, $h_t < \bar{h}$ and $w_t = \overline{w}$. These conditions can be summarized as

$$(w_t - \overline{w})(h_t - \bar{h}) = 0 \tag{3}$$

3.2 Government

The government operates under a fixed exchange rate regime, chooses issuances of long-term bonds, and provides lump-sum transfers to households.

Debt structure. The government issues non-contingent bonds to international lenders. Following Hatchondo and Martinez (2009), each bond issued at date t promises to pay $(1-\delta)\delta^s$ units of the consumption good at date $t+s+1$ for $s = 0, 1, 2, \ldots$, where $\delta \in [0,1]$ is the coupon decay parameter. The Macaulay duration of this payment stream is $D = (1-\delta)^{-1}$. When $\delta = 0$, bonds are one-period; as $\delta \to 1$, they approach perpetuities. Thanks to the recursive coupon structure, the evolution of the face value of outstanding debt can be represented by the following law of motion:

$$b_{t+1} = \delta\, b_t + i_t, \tag{4}$$

where i_t denotes new bond issuances in period t and $\delta\, b_t$ is the stock of debt that remains outstanding after coupon payments $(1-\delta)b_t$ are made.

[1] The CFA franc peg is maintained by France's guarantee of convertibility, conditional on member states maintaining adequate reserves and fiscal discipline.

Government budget constraint. The government finances lump-sum transfers to households T_t using revenue from bond issuances. Letting $d_t \in \{0,1\}$ denote the default decision, with $d_t = 1$ indicating default, the government budget constraint is

$$T_t = \begin{cases} q_t\, i_t - (1-\delta)\, b_t & \text{if } d_t = 0 \quad \text{(repayment)}, \\ 0 & \text{if } d_t = 1 \quad \text{(default)}, \end{cases} \tag{5}$$

where q_t denotes the price at which the government issues new bonds in period t. Under repayment, the government receives revenue $q_t\, i_t$ from new issuances and pays the maturing coupon $(1-\delta)b_t$. Using the law of motion (4) to substitute $i_t = b_{t+1} - \delta\, b_t$, the budget constraint under repayment can equivalently be written as

$$T_t = q_t\, [b_{t+1} - \delta\, b_t] - (1-\delta)\, b_t. \tag{6}$$

Upon default, the government repudiates all obligations and cannot issue new debt in the default period, so transfers are zero.

Hidden debt revelation. The model economy begins at $t = 0$ immediately following the revelation of hidden liabilities. The previous administration accumulated obligations $h \geq 0$ that were excluded from officially reported debt statistics. An audit by the incoming government discloses these liabilities, and they must be recognized. The initial state is therefore $(b_0 + h,\, y_0)$, where b_0 is the pre-revelation official debt stock and h is the revealed hidden debt.

3.3 International Lenders

A continuum of competitive, risk-neutral international lenders can borrow and lend at a constant gross risk-free rate $R^f = 1 + r^f$. Each lender is small and takes the bond price schedule as given. Competition drives lenders' expected returns to the risk-free rate, so bond prices reflect the probability-weighted expected payoffs discounted at R^f.

The standard asset pricing condition for bonds requires that the price at which the government issues debt in period t satisfies

$$q_t = \frac{1}{R^f}\, \mathbb{E}_t \Big\{ (1 - \hat{d}_{t+1})\, [(1-\delta) + \delta\, q_{t+1}] + \hat{d}_{t+1}\, \kappa_{t+1} \left[(1-\delta) + \delta\, q^0_{t+1} \right] \Big\}, \tag{7}$$

where $\hat{d}_{t+1}$ is the equilibrium default decision in period $t+1$, and $q_{t+1} = q(b_{t+2}, y^T_{t+1})$ is the equilibrium bond price in $t+1$ conditional on repayment, where b_{t+2} denoting the government's borrowing decision at $t+1$ and with $q^0_{t+1} = q(0, y^T_{t+1})$. The first term inside the expectation captures the payoff when the government repays in $t+1$: the bondholder receives the current coupon $(1-\delta)$ plus the market value of the remaining bond $\delta\, q_{t+1}$. The second term inside the expectation captures the payoff in default: a fraction κ_{t+1} of the bond value is recovered.

Recovery rates. Following Cruces and Trebesch (2013) and Benjamin and Wright (2018), recovery rates are decreasing in the debt burden and income at the time of default. In period $t+1$, conditional on default, the recovery rate is given by

$$\kappa_{t+1} = \max\left\{\kappa_{\min},\ \kappa_{\max} - \xi_b \frac{b_{t+1}}{y^T_{t+1}} - \xi_y \max\{y^T_{t+1} - \bar{y}^T,\, 0\}\right\}. \tag{8}$$

The parameter $\xi_b > 0$ captures the empirical regularity that higher debt burdens lead to larger haircuts, while $\xi_y > 0$ reflects the finding that defaults during good times are associated with larger creditor losses. For the baseline calibration, we use a flat recovery rate $\kappa = 0.50$, broadly consistent with recent restructurings in Africa.[2] We also report results under the endogenous specification (8) as a robustness check. The sovereign spread is defined as the excess yield on sovereign bonds over the risk-free rate:

$$s_t = \frac{1}{q_t} - R^f. \tag{9}$$

Because international lenders are risk-neutral, variations in spreads are only attributable to default risk and recovery rates.

3.4 Equilibrium

In equilibrium, market-clearing for non-tradable goods requires that output is consumed domestically:

$$c^N_t = y^N_t = h_t. \tag{10}$$

Using the households' optimality condition equating the marginal rate of substitution to the relative price of non-tradables, the firms' optimality condition ($p^N_t = w_t$ under linear technology), the non-tradable goods market clearing condition (10), and the

[2] Ghana's 2024 external bond restructuring involved a 37% nominal haircut with corresponds to a recovery rate $\kappa = 0.63$; Zambia's post-default restructuring delivered materially lower nominal losses than 50%, with estimates varying by instrument and method. The baseline recovery rate of $\kappa = 0.50$ represents the most conservative assumption for our analysis: higher recovery would make lenders more willing to extend credit, which lowers the endogenous borrowing limit and reduces the default threshold. Therefore, our results provide a lower bound on Senegal's default probability.

wage rigidity constraint $w_t \leq \overline{w}$, we can derive the demand for non-tradable goods[3]

$$y_t^N \leq \left(\frac{1-\alpha}{\alpha\,\overline{w}}\right)^{\gamma} c_t^T. \tag{11}$$

which states that employment and thus non-tradable output is an increasing function of tradable consumption when the economy is not at full employment. We can now define a competitive equilibrium for given government policies.

Definition 1 (Competitive equilibrium given policy)**.** *Given initial debt $\{b_0 + h\}$ and set of government policies $\{T_t, b_{t+1}, d_t\}_{t=0}^{\infty}$, a competitive equilibrium is a sequence of allocations $\{c_t^T, c_t^N, h_t\}_{t=0}^{\infty}$ and prices $\{p_t^N, w_t, q_t\}_{t=0}^{\infty}$ such that:*

(i) $\{c_t^T, c_t^N, h_t\}_{t=0}^{\infty}$ *solve households' and firms' problems at given prices;*

(ii) government policies $\{T_t, b_{t+1}, d_t\}_{t=0}^{\infty}$ *satisfy the government budget constraint* (5)*;*

(iii) the bond pricing equation (7) *holds;*

(iv) the market for non-tradable goods clears: $c_t^N = h_t$*; and*

(v) the labor market satisfies: $w_t \geq \overline{w}$, $h_t \leq \bar{h}$ *and the slackness condition* (3)*.*

3.5 Government Problem

In each period, the government chooses between repayment and default. Let $V(b, y^T)$ denote the value function of a government in good credit standing:

$$V\Big(b, y^T\Big) = \max\Big\{V^R\Big(b, y^T\Big),\ V^D\left(y^T\right)\Big\}, \tag{12}$$

where $V^R(b, y^T)$ is the continuation value under repayment and $V^D(y^T)$ is the value of default.

Repayment. A government that repays services its maturing coupons $(1-\delta)b$, chooses end-of-period debt $b' \geq 0$, and the household consumes the residual:

$$V^R(b, y^T) = \max_{b' \geq 0} u\left(c^T(b, b', y^T), y^N\right) + \beta \mathbb{E}_{y^{T'}|y^T} V\left(b', y^{T'}\right) \tag{13}$$

[3]When the wage floor binds, non-tradable output is proportional to tradable consumption, and the CES aggregate reduces to a constant multiple of c^T cT. Under CRRA preferences this rescaling does not alter the government's policy rankings, so the default and borrowing rules are isomorphic to a one-good model at the calibrated parameters. The nominal rigidity nonetheless affects welfare levels and the mapping from debt to consumption, and it disciplines the output cost of default in a monetary union where the exchange rate channel is closed: a given fiscal contraction generates larger employment losses than under a flexible rate, amplifying the cost of the austerity path and thus lowering the threshold at which default is preferred.

subject to

$$c^T(b, b', y^T) = y^T - (1-\delta)b + q(b', y^T)\,[b' - \delta b], \tag{14}$$

$$y^N \le \left(\frac{1-\alpha}{\alpha\overline{w}}\right)^{\gamma} c^T. \tag{15}$$

where $q(b', y^T)$ is the equilibrium bond price schedule and $b' - \delta b$ is net new issuance. The price q depends on the government's future borrowing and default decisions, so the budget set is itself an equilibrium object.

Default. When a government defaults, it repudiates all outstanding obligations and bears two costs. First, it incurs a loss of tradable output which falls to $y^T - \Phi(y^T)$ with

$$\Phi(y^T) = \varphi_0\, y^T + \varphi_1 \max\{y^T - \bar{y}^T,\, 0\}^2. \tag{16}$$

following (Chatterjee and Eyigungor, 2012). When $\varphi_0 < 0$ and $\varphi_1 > 0$, the cost is zero for low output realizations and rises more than proportionately with output above a threshold $-\varphi_0/\varphi_1$, capturing the asymmetry in default punishment first emphasized by Arellano (2008). This asymmetry plays two distinct roles in our quantitative analysis. First, it disciplines the *level* of equilibrium debt and spreads: because default is severely punished in high-income states, lenders rationally expect the sovereign to repay when output is above trend, which sustains low spreads and encourages aggressive borrowing.[4] When output falls, the punishment weakens, default risk rises, and spreads widen, eventually triggering default if income remains depressed. Second, the curvature parameter d_1 governs the *volatility* of spreads, a moment that proportional cost specifications cannot match. A higher d_1 makes the probability of default more sensitive to output fluctuations, generating the large swings in spreads observed in the data.

Second, there is an exclusion from international credit markets. During exclusion household consumes its reduced tradable endowment, $c^T = y^T - \Phi(y^T)$, and each period the government can regain market access with probability $\theta \in (0, 1)$, re-entering with zero debt. The value of default is therefore

$$V^D(y^T) = u\left(y^T - \Phi(y^T), y^N\right) + \beta \mathbb{E}_{y^{T'}|y^T}\left[\theta\, V(0, y^{T'}) + (1-\theta)\, V^D(y^{T'})\right] \tag{17}$$

[4] This amplification channel is also present in the WAEMU: regional banks hold a large share of their assets in government securities, so default during a credit boom inflicts disproportionate banking-sector losses. Default at high income also signals deep solvency problems, triggering capital flight and threatening the currency peg. Further channels include reputational costs and the risk of exclusion from BCEAO liquidity facilities for violating WAEMU fiscal convergence criteria.

subject to

$$y^N \leq \left(\frac{1-\alpha}{\alpha \overline{w}}\right)^{\gamma} \left[y^T - \Phi(y^T)\right]. \tag{18}$$

The government defaults whenever $V^D(y^T) \geq V^R(b, y^T)$. The *default set* at debt level b collects the endowment realizations for which default is weakly preferred:

$$D(b) = \{y^T \in \mathcal{Y} : V^D(y^T) \geq V^R(b, y^T)\}. \tag{19}$$

4 Quantitative Analysis

4.1 Calibration

We solve the model and calibrate parameters to match Senegal's sovereign spread dynamics before and after the hidden debt revelation. The model period is one quarter. Table 1 reports all parameter values.

Table 1: Calibrated Parameters

Parameter	Symbol	Baseline	No-default	Source
Preferences				
Risk aversion	σ	2.0	2.0	Standard
Discount factor	β	0.96	0.96	Standard
CES tradable weight	α	0.26	0.58	SGU / Senegal structure
CES elasticity (T/N)	γ	0.44	0.44	Schmitt-Grohé and Uribe (2016)
Debt structure				
Coupon decay	δ	0.30	0.30	Calibrated
Risk-free rate	r^f	0.01	0.01	U.S. Treasury (quarterly)
Default costs				
Re-entry probability	θ	0.10	0.10	Gelos et al. (2011)
Linear output cost	φ_0	0.091	0.092	Calibrated
Quadratic output cost	φ_1	2.0	2.0	Borensztein and Panizza (2009)
Tradable income process				
Persistence	ρ_y	0.957	0.60	Estimated / market-implied
Innovation std. dev.	σ_y	0.025	0.10	Estimated / market-implied
Recovery				
Flat recovery rate	κ	0.50	0.50	Africa benchmark
Initial conditions (Senegal)				
Pre-revelation b/y	b_0	0.75		Official statistics (pre-audit)
Hidden debt	h	0.45		Court of Accounts audit
Corrected b/y (2025)		1.23		Revised statistics

In our baseline calibration, we target the pre-revelation EMBIG spread at the officially reported debt level $b/y = 0.75$.[5] This calibration yields an output cost $\varphi_0 = 0.091$ and delivers a model-implied spread at reported debt and median income of about 300 basis points, consistent with the 2021–2022 Senegal EMBIG. We then evaluate what happens at the retroactively corrected debt levels, which incorporate the hidden liabilities that were accumulating throughout this period, and deliver a prediction about when default would have been optimal. The remaining parameters are set as follows. The quadratic output cost is set to $\varphi_1 = 2.0$, consistent with the amplified costs of default within a monetary union where banking sector contagion and loss of regional central bank facilities compound the direct trade disruption channels documented in Borensztein and Panizza (2009).

The re-entry probability $\theta = 0.10$ implies an average exclusion duration of 10 quarters, at the upper end of the estimates in Gelos et al. (2011). The CES aggregatiom parameters governing the tradable/non-tradable aggregation ($\alpha = 0.26$, $\gamma = 0.44$) and the wage floor are taken from Schmitt-Grohé and Uribe (2016). The flat recovery rate $\kappa = 0.50$ is broadly consistent with recent restructurings in Africa.

Income process. The tradable endowment follows a stationary first-order autoregressive process:

$$y^T_{t+1} = \mu_y(1-\rho_y) + \rho_y y^T_t + \sigma_y \varepsilon_{t+1}, \qquad \varepsilon_{t+1} \sim \mathcal{N}(0,1), \tag{20}$$

We estimate the parameters (ρ_y, σ_y) from the World Development Indicators. Tradable value added is measured as the sum of agriculture, forestry, and fishing and industry including construction over the period 1980–2024. The estimates are $\rho_y = 0.957$ and $\sigma_y = 0.025$.

4.2 Results

The baseline model predicts default. The baseline calibration, which matches the pre-revelation EMBIG spread at reported debt using the estimated income process, yields a default threshold of $b^* = 1.068$ at median income. Figure 3 plots the repayment and default value functions. The threshold falls well below the retroactively corrected debt levels that prevailed from 2023 onward, implying that default would have been the optimal policy had the true debt stock been known at the time.

Table 2 traces the model's implications year by year. At the officially reported

[5] Hidden liabilities span external PPG debt, off-budget commercial bank loans, special deposit accounts, and an omitted sukuk, while the EMBIG prices only the external bond stock. To the extent that domestic hidden liabilities are not priced by the EMBIG, our spread targets understate the full sovereign risk premium, which would make our default risk assessment conservative.

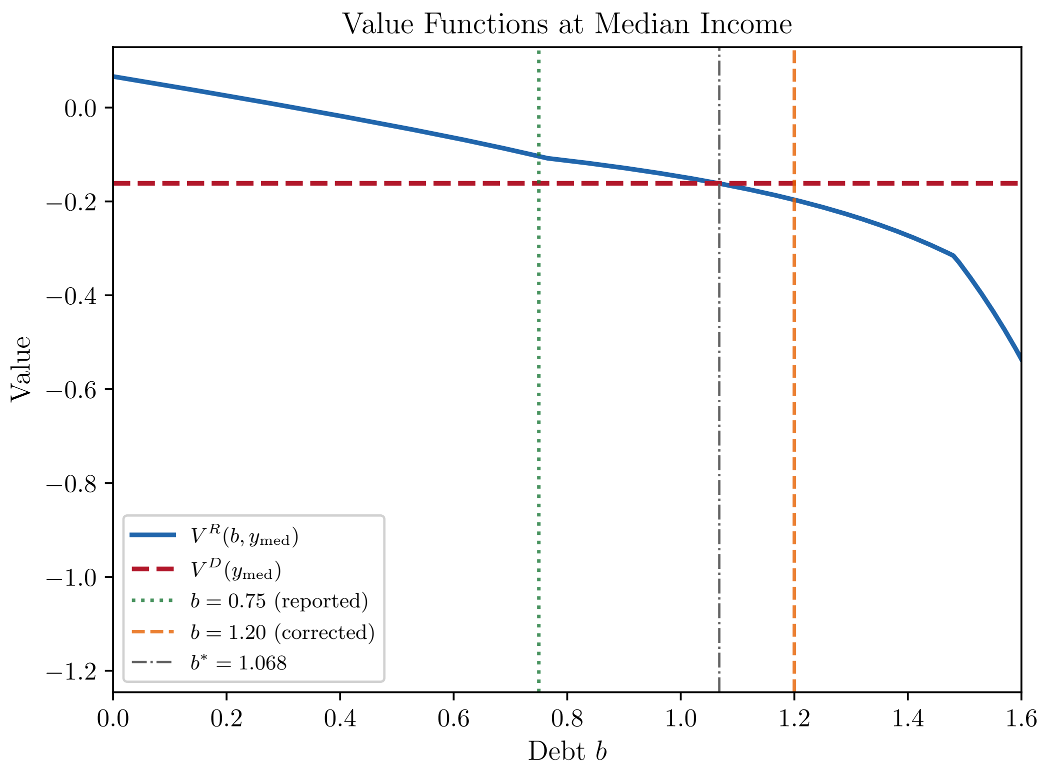


Figure 3: Repayment value $V^R(b, y_{med})$ and default value $V^D(y_{med})$ in the baseline calibration. The default threshold is $b^* = 1.068$. Vertical lines mark officially reported debt ($b/y = 0.75$) and corrected post-revelation debt ($b/y = 1.20$).

debt levels (61 to75 percent of GDP during 2018–2022), the model generates modest spreads of 44–301 basis points, broadly consistent with the pre-revelation EMBIG. At the retroactively corrected debt levels—which incorporate the hidden liabilities accumulating throughout this period—the model implies distressed spreads and, from 2023 onward, outright default. The corrected debt-to-GDP ratio crossed the default threshold of 107 percent between 2022 and 2023, before the audit was conducted and the IMF program was suspended.

Figure 4 displays the default set in the (b, y^T) plane. The staircase-shaped boundary reflects the countercyclical nature of default: lower income realizations expand the set of debt levels at which default is optimal. At the reported debt of 75 percent of GDP (white dashed line), Senegal lies safely in the repayment region. At the corrected post-revelation debt of 120 percent (yellow dashed line), Senegal falls deep inside the default region for all income states at or below the median.

The baseline result is stark: the model predicts that a government that knew its true debt stock should have defaulted from 2023 onward. This conclusion is robust to the recovery rate assumption: under $\kappa = 0.40$, the default threshold rises modestly to

Table 2: Baseline Model: Reported vs Corrected Debt

Year	Debt/GDP (%)		EMBIG	Model spread (bp)		Default
	Reported	Corrected	(bp)	At reported	At corrected	at corr.
2018	61	62	—	44	44	No
2019	64	82	—	45	1,487	No
2020	69	90	—	74	1,780	No
2021	73	99	477	301	1,780	No
2022	75	105	600	301	1,782	No
2023	—	118	561	—	>10,000	**Yes**
2024	—	128	512	—	>10,000	**Yes**

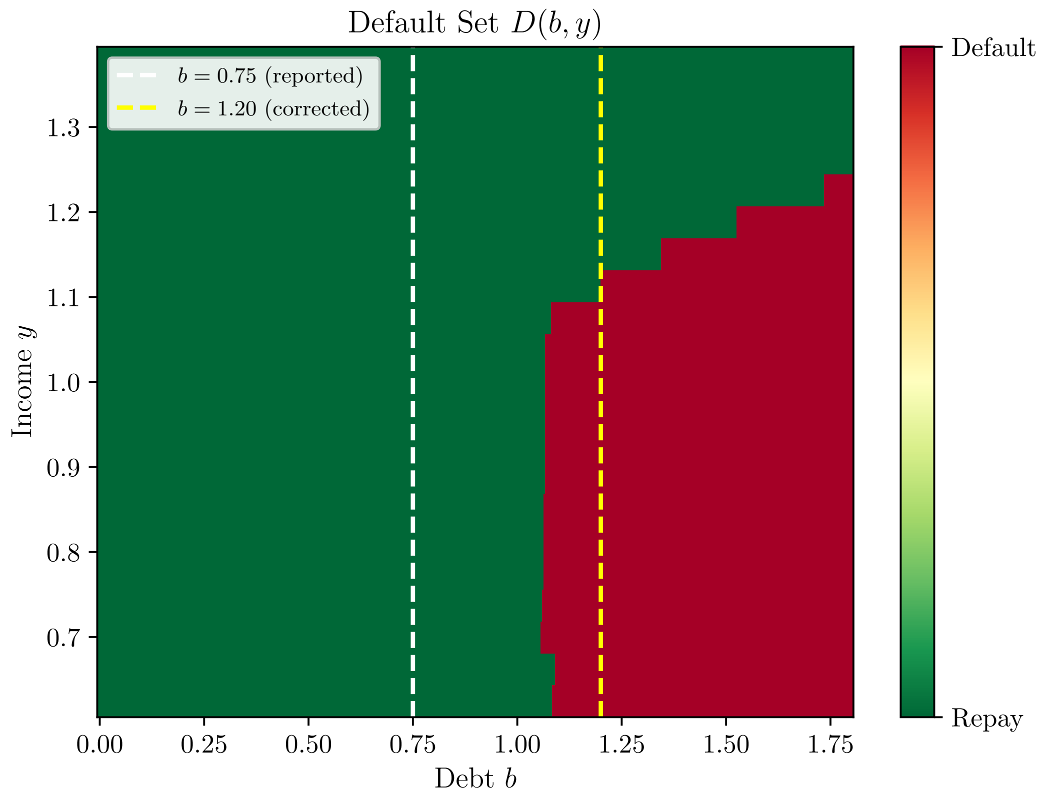


Figure 4: Default set in the baseline calibration. Green: repayment. Red: default. White dashed: reported debt $b/y = 0.75$. Yellow dashed: corrected debt $b/y = 1.20$. The default threshold at median income is $b^* = 1.068$.

$b^* = 1.124$as lenders price in larger restructuring losses, but default remains optimal at all corrected debt levels from 2023 onward.

Rationalizing non-default. Senegal has not defaulted. Through late 2025 and into 2026, the authorities have continued servicing external obligations and have signaled their intention to avoid restructuring. To study the stability of this outcome, we consider a no-default calibration in which the model rationalizes continued repayment and matches the observed spread dynamics.

The alternative calibration has two key elements. The first is the income-change identification strategy. Because the hidden liabilities existed long before the audit, using the reported debt level $b/y = 0.75$ has its limits, as Senegal's EMBIG spread could reflect non-public information on the hidden debt in the 2019-2024 period. Therefore, we evaluate the model's spread schedule at the true debt stock and use variation in real income across two dates to identify an alternative default region that rationalizes Senegal not defaulting at the end of 2025/beginning of 2026. We target two EMBIG moments, both evaluated at the 2025 corrected debt-to-GDP ratio of 123 percent, but at different income states: August 2022 (spread $\approx$ 675 basis points) at income 5.6 percent above the median, and November 2025 (spread $\approx$ 1,150 basis points) at median income.[6] The 5.6 percent income decline between the two dates reflects the cumulative deterioration from below-trend growth, fiscal consolidation, and the contractionary feedback from the revelation.[7] Both moments are read off the same equilibrium spread schedule, disciplined by the output cost φ_0 and coupon decay δ.

The second element is income optimism. The baseline income process, estimated from WDI data on Senegal's tradable sector, has quarterly persistence $\rho_y = 0.957$ and innovation standard deviation $\sigma_y = 0.025$. Under these parameters, quarterly income transitions are too concentrated to generate the intermediate spread levels (500–1,200 basis points) observed in the EMBIG. Matching the spread dynamics requires an income process with lower effective persistence ($\rho_y = 0.60$) and correspondingly higher innovation volatility ($\sigma_y = 0.10$).

The gap between the estimated income persistence ($\rho_y = 0.957, \sigma_y = 0.025$) and the market-implied persistence in the No-default calibration ($\rho_y = 0.60, \sigma_y = 0.1$) that rationalizes non-default admits two complementary interpretations. The first is that the market prices additional sources of risk (higher variance) that the production-side AR(1) does not capture: terms-of-trade volatility for a commodity-dependent economy, fiscal uncertainty following the revelation, and the possibility of further undisclosed li-

[6] Note that the corrected debt stock in August 2022 was approximately 105 percent of GDP, not the 2025 level of 123 percent that we use as the fixed debt state. We evaluate both moments at 123 percent for two reasons. First, the 2025 corrected stock is the most precisely measured since it is the final outcome of the audit process. Second, using 123 percent for both moments is the more conservative assumption for the no-default calibration: it requires the model to rationalize a relatively modest spread of 675 basis points at a high debt level, which pushes against finding fragility rather than toward it.

[7] The 5.6 percent income difference corresponds to the cumulative shortfall of Senegal's GDP relative to its pre-COVID trend growth of approximately 6.52 percent per year (2014–2018 average). GDP growth averaged 3.9 percent in 2022, 4.3 percent in 2023, and 6.1 percent in 2024. We were unable to find a definitive estimate of Senegal's 2025 growth rate. The cumulative slow growth in 2022-2024 leads to $\approx$ 5.6 below trend income.

abilities. The second, on the lower persistence, is that the government behaves as if negative income shocks are more transitory than the historical data suggests. That is, it is more optimistic about mean reversion than the econometric estimate warrants. This optimism may reflect genuine expectations about structural improvements, new oil and gas revenues, governance reforms under the incoming administration, alternative sources of financing, or anticipated concessional support from bilateral and multilateral creditors. It may also reflect a form of "gambling for redemption" in which the government delays restructuring in the hope that favorable shocks will restore debt sustainability without the costs of default. Under either interpretation, the lower effective persistence makes the government more willing to endure the short-term pain of debt service at elevated spreads, because it assigns a higher probability to states in which income recovers, and the debt burden becomes manageable.

Under these parameters, the calibration yields $\varphi_0 = 0.092$, $\delta = 0.30$, and a CES tradable weight $\alpha = 0.58$. The higher tradable share relative to the Schmitt-Grohé and Uribe (2016) benchmark ($\alpha = 0.26$) reflects Senegal's economic structure, where agriculture, fishing, mining, and manufacturing account for a larger share of output than in the average emerging market. The No-default calibration generates a default threshold of $b^* = 1.24$ at median income, placing Senegal's post-revelation debt of 123 percent inside the repayment zone but close to the boundary.

The calibrated model matches both target moments exactly: 675 basis points at above-median income and 1,152 basis points at median income, both at $b/y = 1.23$. The spread at $b/y = 1.23$ is highly sensitive to income: it ranges from 675 basis points at $y = 1.056$ (above median) to 1,876 basis points at $y = 0.944$ (below median), with the entire post-revelation EMBIG evolution explained by movement along this single curve as macroeconomic conditions deteriorated.

Table 3 reports the implications of adverse income shocks starting from Senegal's recent state (November 2025, $b/y = 1.23$, median income). At median income, Senegal faces a 19.7% quarterly probability of default. Each percentage point of tradable GDP loss adds approximately 130 basis points to sovereign spreads and raises the quarterly default probability by approximately 2 percentage points. A 20 percent increase in oil prices—costing approximately 4 percent of GDP given Senegal's energy import dependence and subsidy commitments—would raise spreads from 1,152 to approximately 1,670 basis points and increase the default probability from 19.7 to 27.5 percent.[8] Default becomes the optimal policy for income realizations below $y \approx 0.89$, corresponding to an income decline of approximately 11 percent from the current level.

[8] A 20 percent increase in oil prices raises Senegal's fuel import bill by approximately 3 percent of GDP (fuel imports averaged 14–17 percent of GDP during 2022–2024, based on the World Development Indicators series. Including the efficient cost of energy subsidies, the total impact on the government's effective income is approximately 3–4 percent of GDP.

As a robustness check, we lower the recovery rate to $\kappa = 0.40$, recalibrating the no-default model to the same two EMBIG moments. The default threshold is unchanged at $b^* = 1.24$,[9] but spreads at $b/y = 1.23$ rise: 812 basis points at above-median income and 1,404 basis points at median income. The higher spreads reflect the larger losses creditors would incur in a restructuring, which tightens borrowing conditions even when the default decision itself is unchanged.

Table 3: Income Shock Scenarios (No-Default Calibration, Nov 2025, $b/y = 1.23$)

Scenario	Income y	Spread (bp)	ΔSpread (bp)	Default prob. (%)
Current (Nov 2025)	1.000	1,152	—	19.7
−3% tradable GDP	0.970	1,542	+390	25.7
−4% (oil shock, +20%)	0.960	1,670	+518	27.5
−5% tradable GDP	0.950	1,794	+642	29.4
−8% (severe recession)	0.920	2,142	+990	34.1

The two calibrations bracket Senegal's situation. The baseline, anchored to the estimated income process, predicts that default would have been optimal from 2023. The no-default calibration, which rationalizes continued repayment through market-implied income dynamics, places Senegal inside the repayment zone but close to the boundary. Under either interpretation, the hidden debt revelation moved Senegal from a position of moderate fiscal risk into a zone of acute vulnerability where an adverse income shock of 3 percent or more would make restructuring the optimal policy.

4.3 Discussion

The tension between the two calibrations is informative. The baseline, disciplined by the pre-revelation EMBIG and the estimated income process, says default is optimal at the corrected debt levels. The no-default calibration, which matches the post-revelation spread dynamics, requires market-implied income dynamics that are more volatile and less persistent than what the historical production data suggests. Our takeaway from the two calibrations is that Senegal should have either defaulted at the revelation of its hidden liabilities, or is in a fragile equilibrium that any moderate adverse shock could tip over.

[9] In general equilibrium, a lower recovery rate reduces bond prices and tightens borrowing conditions, which lowers the value of repayment and would ordinarily shift the default threshold downward. In our recalibration, however, the output cost φ_0 adjusts upward to re-match the target spread moments, partially offsetting the tightening effect on repayment values; the net result is that the default threshold remains at $b^* = 1.24$.

Several forces outside the model may explain why Senegal has not defaulted despite the baseline prediction: political costs specific to a new administration that promised fiscal rectitude; the strategic value of maintaining market access within WAEMU, where a default by one member could destabilize the regional banking system and the currency peg; and the expectation of concessional support from bilateral or multilateral creditors. In the no-default calibration, these forces are captured implicitly through the lower income persistence, which makes the government more willing to endure short-term pain in expectation of income recovery.

We acknowledge that modeling hidden debt as an initial condition is a deliberate simplification that abstracts from the dynamics of accumulation and discovery. The paper's primary contribution is to characterize the optimal policy response conditional on revelation having occurred, rather than to explain why or when revelation happens. The baseline finding that default would have been optimal from 2023 — before the 2024 audit — should be read as a statement about the corrected debt stock, not as a claim that revelation itself triggers default: it establishes that the hidden liabilities, had they been publicly known, would have placed Senegal in the default region throughout this period. Endogenizing the revelation through a monitoring game, as in Cole and Kehoe (2000), is a natural extension but is not necessary for studying the optimal policy response conditional on the revelation having occurred.

5 Conclusion

This paper studies optimal sovereign default following the revelation of hidden public liabilities, using Senegal's 2024 debt shock as a laboratory.

The baseline calibration, disciplined by pre-revelation EMBIG spreads and the estimated tradable income process, delivers a stark prediction: at the retroactively corrected debt levels, default would have been optimal from 2023 onward — a year before the audit was conducted and the hidden liabilities became public. That Senegal has not defaulted despite this prescription suggests that forces outside the model sustain repayment beyond what the standard calculus of output costs and market exclusion would predict.

A no-default calibration that rationalizes continued repayment requires market-implied income dynamics in which shocks are more transitory and volatile than historical data suggests, reflecting optimism for a reversal (low persistence), terms-of-trade risk, political uncertainty, and the possibility of further hidden liabilities (higher variance). Under this calibration, Senegal is solvent but fragile: the post-revelation debt sits close to the default boundary, and an adverse income shock of 3 percent or more—from a recession, or a sustained increase in oil prices—would lead to an expected time to default of less than one year. Both calibrations point to the same conclusion: the hidden-debt revelation moved Senegal from a position of moderate fiscal risk into a zone of acute vulnerability.

These findings carry a broader lesson for hidden-debt episodes in frontier markets. The revelation of hidden liabilities does not merely raise the level of debt; it can move a sovereign past the point where standard models predict default should occur. When the sovereign nonetheless continues to repay, the resulting high spreads and elevated default probability create a fragile equilibrium that is sustained by expectations of favorable adjustment or support, but that can unravel quickly if those expectations are disappointed. Designing better disclosure frameworks and early-warning systems for hidden liabilities is therefore not only a matter of fiscal transparency; it is a precondition for preserving the policy space that sovereigns need to avoid unnecessary default.

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Appendix

5.1 Computational Algorithm

We solve for the recursive equilibrium by value function iteration. The algorithm proceeds as follows.

Step 0: Discretization. The debt stock is discretized on a grid $\mathcal{B} = \{b_1, \ldots, b_{N_b}\}$ with $b_1 = 0$ and $b_{N_b} = b_{\max}$. We use $N_b = 300$ grid points with a finer mesh on the interval $[0.6, 1.3]$ in order to capture the pre- and post-revelation dynamics accurately. The income process (20) is discretized using the Rouwenhorst method with $N_y = 21$ states.

Step 1: Initialization. The bond price is initialized at the risk-free level: $q^{(0)}(b', y) = (1-\delta)/(R^f - \delta)$, which is the price of a default-free long-term bond with coupon decay δ. The value functions are initialized at $V^{(0)}(b, y^T) = u(y^T)/(1-\beta)$ and $V^{D,(0)}(y^T) = u(y - \Phi(y^T))/(1-\beta)$.

Step 2: Iteration. Given the current iterates $(V^{(n)}, V^{D,(n)}, q^{(n)})$, we update the value functions and the pricing schedule:

(a) *Update default value.* For each y_i:

$$V^{D,(n+1)}(y_i) = u\big(y_i - \Phi(y_i)\big) + \beta \sum_{j=1}^{N_y} \Pi_{ij} \Big[\theta\, V^{(n)}(0, y_j) + (1-\theta)\, V^{D,(n)}(y_j)\Big]. \quad (21)$$

(b) *Update repayment value.* For each (b_k, y_i):

$$V^{R,(n+1)}(b_k, y_i) = \max_{\ell \in \{1,\dots,N_b\}} \left\{ u(c_{ki\ell}) + \beta \sum_{j=1}^{N_y} \Pi_{ij}\, V^{(n)}(b_\ell, y_j) \;:\; c_{ki\ell} \geq 0 \right\}, \quad (22)$$

where $c_{ki\ell} = y_i - (1-\delta)b_k + q^{(n)}(b_\ell, y_i)\big[b_\ell - \delta b_k\big]$. Store the optimal policy $g^{(n+1)}(b_k, y_i) = b_{\ell^*}$. If no ℓ yields $c_{ki\ell} \geq 0$, set $V^{R,(n+1)}(b_k, y_i) = -\infty$.

(c) *Update value function:* $V^{(n+1)}(b_k, y_i) = \max\{V^{R,(n+1)}(b_k, y_i),\ V^{D,(n+1)}(y_i)\}$.

(d) *Update bond price.* Evaluate the right-hand side of equation (7) using the updated value function, policy functions, and default set. We update the price schedule using a dampened iteration, $q^{(n+1)} = a\, \hat{q}^{(n+1)} + (1-a)\, q^{(n)}$ with $a = 0.3$, to improve convergence.

Step 3: Convergence. The algorithm terminates when $\max\{\|V^{(n+1)} - V^{(n)}\|_\infty,\ \|q^{(n+1)} - q^{(n)}\|_\infty\} < \epsilon$, with $\epsilon = 10^{-5}$ for value functions and $\epsilon = 10^{-3}$ for prices.